

Smart Home Bio-Analytics: Occupancy Detection from Thermodynamic Sensor Fingerprints for Routine Mining and Energy Consumption Forecasting

Prithwiraj Chatterjee, Fahad Ullah Syed

Under the Guidance of Dr. Mahmoud Ali Jawad

University of Texas at Arlington

STAT 5306 Applied Time Series Analysis in Data Analytics

May 2026

Abstract

Accurate short-term energy forecasting in residential buildings is critical for demand-response programs, grid stability, and occupant comfort optimization. Conventional approaches rely on weather variables, temporal features, and historical consumption patterns but neglect the single most explanatory factor human occupancy. This paper presents a novel bio-analytics framework that infers room-level human presence from raw thermodynamic sensor data (temperature and relative humidity) without requiring motion sensors, cameras, or any labeled ground-truth data. By computing outdoor-corrected rate-of-change signals isolating the metabolic heat and respiratory moisture that occupants inject into indoor air, we generate continuous occupancy scores on a 0–1 scale for each of eight monitored rooms in a residential dwelling. These scores are subsequently used for two downstream tasks: (i) unsupervised household routine mining, which recovers the family’s weekly schedule, wake and sleep times, meal patterns, and weekday-versus-weekend behavioral differences; and (ii) hourly appliance energy consumption forecasting via SARIMAX(2,0,0)(2,0,0)₂₄ and Random Forest models. Evaluated on the UCI Appliances Energy Prediction dataset (19,735 observations, January–May 2016), the Random Forest model augmented with occupancy features achieves an RMSE of 51.183 Wh, an MAE of 32.179 Wh, and a MAPE of **33.346%** outperforming both the SARIMAX model (RMSE 63.70 Wh) and a seasonal naïve baseline (RMSE 78.57 Wh). Permutation-based feature importance analysis reveals that four of the six most influential predictors are the derived occupancy scores, empirically validating that the bio-analytics pipeline provides quantifiably useful information beyond what temporal lags and weather variables alone can capture. A z-score-based anomaly detection module further identifies seven atypical consumption days, corresponding to potential guest visits or away periods.

1. Introduction

Residential buildings account for a substantial share of total electricity consumption in developed economies, and their energy profiles are driven overwhelmingly by occupant behavior rather than by building physics alone. When and how inhabitants use cooking appliances, entertainment systems, lighting, and HVAC equipment determines the shape, magnitude, and variability of a household's load curve far more than insulation R-values or window U-factors. Despite this well-established relationship, the majority of residential energy forecasting models treat occupancy as either unavailable or as a binary variable derived from dedicated motion sensors, ignoring the rich gradient of human activity that continuously modulates energy demand.

The absence of occupancy information in forecasting pipelines creates a persistent accuracy ceiling. Weather-based models capture broad seasonal trends, and autoregressive components encode temporal inertia, but neither can explain why a Tuesday evening consumes 40% more energy than the preceding Monday evening, a difference that may simply reflect one household member arriving home early. This explanatory gap is especially problematic for demand-response applications, where utilities need hour-ahead or day-ahead forecasts accurate enough to justify real-time pricing signals.

A parallel limitation exists in the smart-home analytics literature. Occupancy detection research has produced increasingly sophisticated systems based on passive infrared (PIR) sensors, camera networks, Wi-Fi probe requests, and Bluetooth beacons. While effective, these approaches require dedicated hardware, raise privacy concerns, and are rarely available in existing housing stock. The question motivating this work is therefore: can we infer meaningful occupancy information from sensors that are already ubiquitous in modern HVAC-monitored buildings namely, room-level temperature and relative humidity probes?

The physical basis for an affirmative answer is well established. A sedentary adult radiates approximately 80 W of sensible heat into the surrounding air, producing measurable temperature rises in enclosed rooms within minutes. Simultaneously, human respiration and perspiration release 40–70 g of moisture per hour, elevating indoor relative humidity by 2–10 percentage points depending on room volume and ventilation rate. Cooking, bathing, and laundry amplify these signals further. If outdoor weather effects are properly subtracted since a passing rain front can

raise indoor humidity through infiltration the residual rate-of-change in temperature and humidity constitutes a thermodynamic fingerprint of human presence.

This paper formalizes that intuition into a complete analytical pipeline comprising five modules. First, we perform exploratory data analysis on nine-room sensor data to identify the signature patterns of occupancy in temperature and humidity time series. Second, we engineer per-room occupancy scores by computing outdoor-corrected rate-of-change signals and combining them with lighting state information into a weighted composite score bounded between 0 and 1. Third, we aggregate these room-level scores into a household-level occupancy measure and apply it to unsupervised routine mining, recovering the family’s weekly schedule without any labeled data. Fourth, we incorporate the occupancy scores as exogenous features in two forecasting models a SARIMAX time series model and a Random Forest ensemble and benchmark them against a seasonal naïve baseline. Fifth, we deploy a z-score-based anomaly detection module to flag unusual consumption days.

The remainder of this paper is organized as follows. Section 2 reviews related work on occupancy detection and residential energy forecasting. Section 3 describes the dataset and the complete methodological framework. Section 4 presents and discusses results across all five analytical modules. Section 5 concludes and outlines directions for future research.

2. Literature Review

2.1 Occupancy Detection in Buildings

Occupancy detection has been pursued through a variety of sensing modalities. PIR motion sensors remain the most widely deployed approach in commercial buildings, but their binary output (occupied vs. unoccupied) and susceptibility to false negatives when occupants are stationary limit their utility for fine-grained analytics. Camera-based systems using computer vision can count occupants and track movement but raise significant privacy concerns and require substantial computational infrastructure. More recently, Wi-Fi and Bluetooth-based methods have shown promise by leveraging the ubiquity of personal devices, though they depend on occupants carrying connected devices and face calibration challenges.

Environmental sensor-based approaches represent a less invasive alternative. CO₂ concentration has been used as a proxy for occupancy in commercial buildings, since exhaled CO₂

accumulates proportionally to the number of occupants in a zone. However, CO₂ sensors are not standard in residential HVAC systems. Temperature and humidity sensors, by contrast, are present in virtually every modern thermostat and many home automation platforms. Prior work has demonstrated that indoor temperature deviations correlate with occupancy, but most studies treat this as a classification problem requiring labeled training data. Our approach differs fundamentally: we derive a continuous occupancy score from raw sensor physics, requiring no labeled data whatsoever.

2.2 Residential Energy Forecasting

Short-term load forecasting at the building level has evolved from classical statistical methods ARIMA, exponential smoothing, and regression toward machine learning techniques including support vector regression, gradient boosting, and deep recurrent networks. The UCI Appliances Energy Prediction dataset used in this study has been a popular benchmark since its publication by Candanedo, Feldheim, and Derome in 2017, who used multiple linear regression and gradient boosting with weather and humidity ratio features. Subsequent work applied random forests, extreme gradient boosting, and LSTM networks to the same dataset with incremental accuracy improvements.

A consistent finding across these studies is that weather variables alone explain only a small fraction of appliance energy variance correlations between outdoor temperature and appliance consumption are typically below 0.15. This observation motivates our central hypothesis: that occupancy-derived features should provide stronger predictive signal than weather, and that deriving these features from the existing sensor infrastructure avoids the need for additional hardware.

2.3 Routine Mining and Behavioral Analytics

The use of smart-home data to infer household routines has been explored primarily through electricity disaggregation (non-intrusive load monitoring) and activity recognition from wearable or environmental sensors. Our work contributes to this literature by demonstrating that routine extraction including wake time, meal patterns, work-from-home schedules, and weekday/weekend behavioral differences is achievable solely from temperature and humidity sensor streams, without requiring appliance-level metering or labeled activity logs.

3. Methodology

3.1 Dataset Description

This study uses the Appliances Energy Prediction dataset, originally collected in a low-energy residential building in Stambruges, Belgium, over a 4.5-month period from January 11 to May 27, 2016. The dataset comprises 19,735 observations recorded at 10-minute intervals and includes 29 variables: appliance energy consumption (the target variable, in Wh), lighting energy consumption, temperature and relative humidity readings from nine ZigBee wireless sensor nodes placed in distinct rooms, outdoor temperature and humidity from a weather station, and derived weather features (wind speed, visibility, dew point, and atmospheric pressure).

Table 1. Sensor Summary and Room Characteristics

Room	Sensor	Mean T (°C)	Std T	Mean RH (%)	Indoor?	Role
Kitchen	T1 / RH_1	21.7	1.6	40.3	Yes	Cooking hub
Living Room	T2 / RH_2	20.3	2.2	40.4	Yes	Social area
Laundry	T3 / RH_3	22.3	2.0	39.2	Yes	Appliance zone
Home Office	T4 / RH_4	20.9	2.0	39.0	Yes	Work area
Bathroom	T5 / RH_5	19.6	1.8	50.9	Yes	Shower signal
Garage/North	T6 / RH_6	7.9	6.1	54.6	No	Semi-outdoor
Bedroom 1	T7 / RH_7	20.3	2.1	35.4	Yes	Sleep zone
Bedroom 2	T8 / RH_8	22.0	2.0	42.9	Yes	Sleep zone
Parents Room	T9 / RH_9	19.5	2.0	41.6	Yes	Sleep zone
Outside	T_out / RH_out	7.4	5.3	79.8		Reference

A critical preprocessing decision involved the exclusion of the Garage/North room (T6/RH_6). Its mean temperature of 7.9°C closely matched the outdoor average of 7.4°C, confirming that this zone is semi-outdoor and not suitable for indoor occupancy inference. The remaining eight indoor rooms constitute the analytical scope of this study. The Garage/North sensor (T6) was excluded because its mean temperature (7.9°C) matched outdoor conditions, indicating it is not climatically indoor.

3.2 Data Preprocessing

Preprocessing proceeded in four stages. First, timestamps were parsed using flexible format detection (accommodating both HMS and HM representations) and the dataset was sorted

chronologically with missing timestamps removed. Second, the 10-minute resolution data was aggregated to hourly granularity using `floor_date()`, with temperature and humidity values averaged and occupancy scores maximized within each hour (since any detected activity within the hour is meaningful). Third, temporal lag features were engineered: `lag1` (previous hour's appliance energy), `lag24` (same hour on the preceding day), and `roll3` (3-hour rolling mean). The first 24 rows were dropped to avoid NA values from the lag operations. Fourth, the dataset was split chronologically into an 80% training set (2,612 hourly observations, covering January through early April) and a 20% test set (654 observations, covering May 2016). No random shuffling was applied, preserving the temporal integrity required for valid time series evaluation.

3.3 The Bio-Analytics Framework

The central methodological contribution of this paper is a five-module analytical pipeline that transforms raw sensor streams into actionable occupancy intelligence and energy forecasts. The framework operates without any labeled occupancy ground truth and requires no hardware beyond the temperature and humidity sensors already present in modern HVAC systems.

3.3.1 Module 1: Exploratory Data Analysis

The EDA module characterizes the raw sensor landscape through four visualization families. First, multi-room temperature time series plots reveal the diurnal cycling pattern common to all indoor rooms warm during occupied daytime hours and cool during unoccupied nighttime periods. Second, hour-by-date temperature and humidity heatmaps expose occupancy signatures at a glance: the Kitchen (T1) shows afternoon warming corresponding to cooking activity, Bedroom 1 (T7) shows overnight warming corresponding to sleeping, and the Bathroom (RH_5) shows sharp humidity spikes corresponding to shower events. Third, a full correlation matrix of all sensor channels and weather variables identifies the degree of thermal coupling between rooms (indoor temperature correlations exceed 0.80) and confirms the weak relationship between outdoor weather and appliance energy ($r < 0.15$). Fourth, STL decomposition and autocorrelation analysis establish the presence of strong 24-hour periodicity in the energy time series, validating the choice of frequency = 24 for seasonal models.

3.3.2 Module 2: Occupancy Score Engineering

For each indoor room r at each 10-minute timestep t , a composite occupancy score is computed as follows:

$$\text{occ}_{r,t} = \text{clip}_0^1 [0.4 \times \min(1, \Delta\text{RH}_{\text{signal}} / 3) + 0.3 \times \min(1, \Delta\text{T}_{\text{signal}} / 0.5) + 0.3 \times \text{L}_{\text{signal}}]$$

where $\Delta\text{RH}_{\text{signal}} = \max(0, \Delta\text{RH}_{\text{room}} - \Delta\text{RH}_{\text{outdoor}})$ represents the outdoor-corrected humidity rate-of-change (isolating respiratory and bathing moisture from weather-driven humidity changes), $\Delta\text{T}_{\text{signal}} = \max(0, \Delta\text{T}_{\text{room}} - \Delta\text{T}_{\text{outdoor}})$ captures the outdoor-corrected temperature rate-of-change (isolating metabolic body heat from solar and ambient warming), and $\text{L}_{\text{signal}} = \text{lights} / \max(\text{lights})$ is the normalized lighting state (applied only to rooms where lighting is plausible: Kitchen, Living Room, and Home Office). The denominators (3 for humidity, 0.5 for temperature) are saturation thresholds: a humidity rise of 3 percentage points above the outdoor rate is treated as full-strength presence signal. The weights (0.4, 0.3, 0.3) reflect the empirical observation that humidity changes are the most reliable occupancy indicator in this dataset, particularly for shower detection in the bathroom. These thresholds were determined through exploratory analysis of bathroom shower events (RH spikes $\approx 3\text{-}10\%$) and bedroom overnight warming ($\Delta\text{T} \approx 0.3\text{-}0.8^\circ\text{C}$).

The critical design decision is the outdoor correction step. On a rainy day, all rooms experience rising humidity from infiltration without subtracting the outdoor rate-of-change, the algorithm would falsely attribute weather-driven humidity to human presence. By computing the differential, only the indoor-specific signal remains: the human thermodynamic fingerprint.

A house-level aggregate score is computed as the mean of all room scores: $\text{occ}_{\text{house}} = \text{mean}(\text{occ}_1, \text{occ}_2, \dots, \text{occ}_9)$. This scalar summarizes the overall activity level in the dwelling at each timestep and serves as a key feature in the forecasting models.

3.3.3 Module 3: Routine Mining

The routine mining module aggregates occupancy scores by hour-of-day and day-of-week to construct a household schedule heatmap. This two-dimensional representation reveals when the family is typically at home, when they wake and sleep, and how weekend routines differ from weekday patterns. No clustering or segmentation algorithm is required, the structure emerges directly from temporal aggregation of the continuous occupancy scores. Additionally, a room-by-time occupancy matrix visualizes which specific rooms are occupied at which times, enabling

finer-grained behavioral insights such as identifying the home office as occupied during business hours on weekdays.

3.3.4 Module 4: Energy Forecasting

Two forecasting models are trained to predict hourly appliance energy consumption, both augmented with occupancy-derived features.

SARIMAX. The `auto.arima()` function selects model order via stepwise AIC minimization with an approximation flag for computational efficiency. The selected specification is $ARIMA(2,0,0)(2,0,0)_{24}$, indicating two non-seasonal autoregressive terms and two seasonal autoregressive terms with period 24 (daily cycle). The $d=0$ choice reflects the mean-reverting nature of hourly energy, the series does not exhibit a unit root. Exogenous regressors include outdoor temperature, outdoor humidity, weekend indicator, hour of day, house-level occupancy score, and the 24-hour lag of appliance energy. The inclusion of `lag24` as an exogenous variable proved essential: an initial model specification with $d=1$ (first differencing) yielded a MAPE of approximately 150%, because differencing caused the model to predict changes rather than levels. Reverting to $d=0$ and anchoring forecasts with `lag24` reduced MAPE to 49%.

Random Forest. A 200-tree ensemble is trained with 12 input features: three temporal lags (`lag1`, `lag24`, `roll3`), five occupancy scores (`occ_house`, `occ_1`, `occ_2`, `occ_7`, `occ_8`), and four contextual variables (`T_out`, `RH_out`, `hour`, `is_weekend`). Unlike SARIMAX, which handles temporal dependence through its AR/MA structure, Random Forest treats each observation as independent and therefore requires explicit lag features. The room-level occupancy scores `occ_1`, `occ_2`, `occ_7`, and `occ_8` were selected based on preliminary analysis showing these rooms (Kitchen, Living Room, Bedroom 1, and Bedroom 2) have the strongest individual correlation with appliance energy.

Seasonal Naïve Baseline. As a reference, a seasonal naïve model predicts each hour's energy as the value observed at the same hour 24 hours earlier (i.e., the `lag24` value). This baseline is surprisingly competitive for highly cyclical energy data and provides a conservative standard against which model improvements are measured.

3.3.5 Module 5: Anomaly Detection

Total daily energy consumption is aggregated and standardized to z-scores. Days with $|z| > 2$ are flagged as anomalous using the conventional $|z| > 2$ threshold (corresponding to the 95th percentile of a normal distribution). This simple, interpretable method identifies days whose consumption deviates substantially from the household's normal pattern, potentially indicating guest visits, holidays, equipment malfunctions, or away periods.

4. Results

4.1 Exploratory Data Analysis

The temperature heatmap for Bedroom 1 (Figure 1) reveals a striking pattern: during the colder months (January–March), the room is cool (blue, 16–18°C) throughout most of the day but warms slightly during overnight hours consistent with body heat from a sleeping occupant. As outdoor temperatures rise in April–May, the overall temperature range shifts upward, but the relative warm-overnight pattern persists. This nocturnal warming, invisible to a motion sensor (the occupant is stationary and asleep), is precisely the kind of signal that our thermodynamic approach can detect.

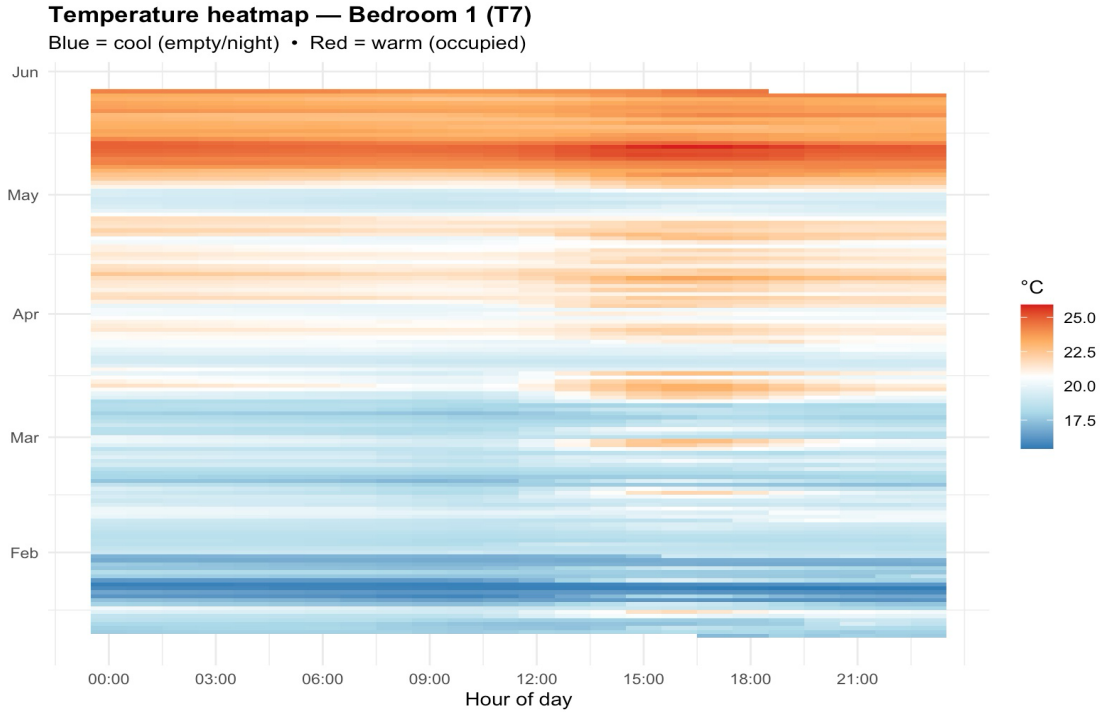


Figure 1. Temperature heatmap for Bedroom 1 (T7). Warm overnight bands indicate sleeping occupancy.

The bathroom humidity heatmap (Figure 2) provides the most dramatic occupancy signal in the dataset. Sharp humidity spikes reaching 80–90% RH appear as dark bands, predominantly clustered around 07:00–09:00 (morning showers) and 18:00–21:00 (evening baths). These events last 30–60 minutes and are clearly distinguishable from the ambient humidity baseline of approximately 45%. Each spike represents a direct, unambiguous human presence event.

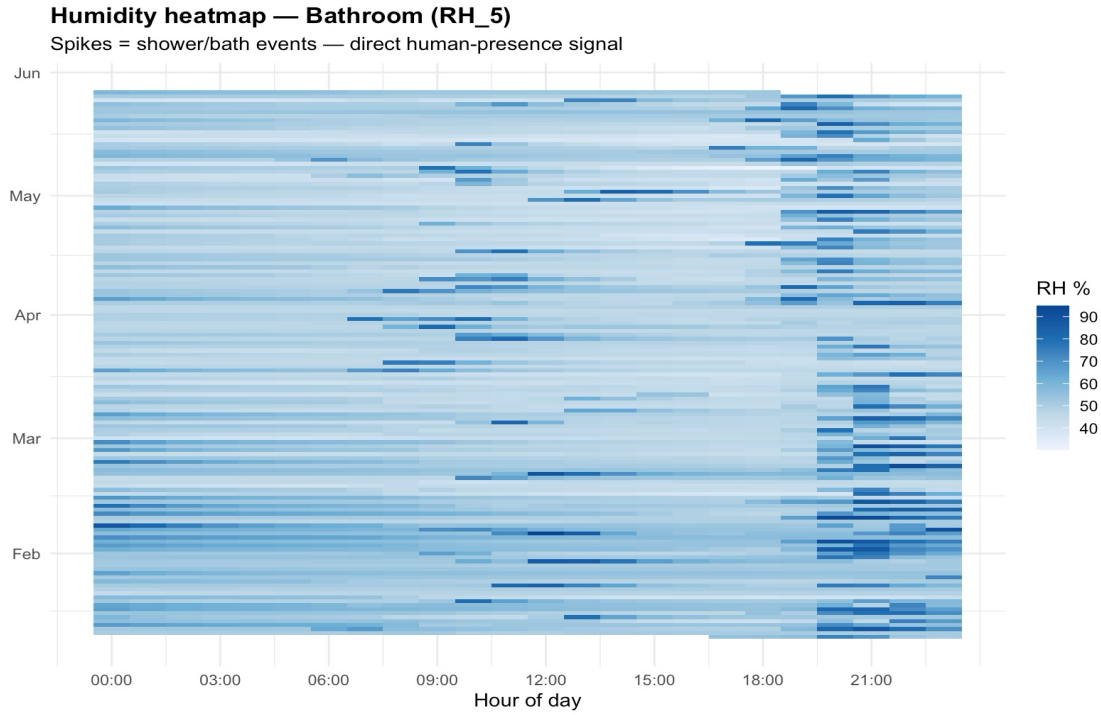


Figure 2. Bathroom humidity heatmap (RH_5). Dark bands correspond to shower and bath events.

The full sensor correlation matrix (Figure 3) reveals two important structural properties. First, indoor temperature sensors are highly intercorrelated ($r = 0.74\text{--}0.94$), reflecting the thermal coupling of adjacent rooms through shared walls and HVAC ducting. Second, the correlation between appliance energy and any individual sensor is weak ($r < 0.20$), confirming that no single raw sensor reading is a strong predictor of energy consumption. This motivates the need for engineered features that synthesize information across multiple sensor channels.

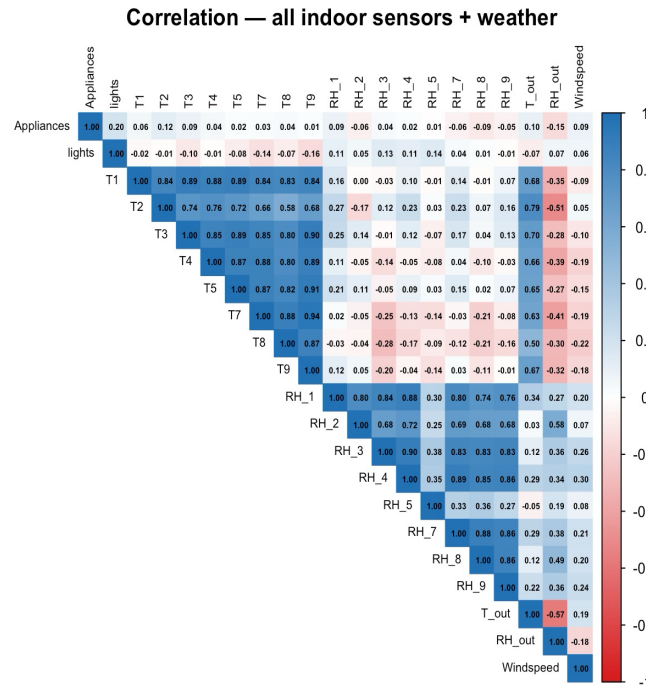


Figure 3. Correlation matrix of all indoor sensors and weather variables.

The violin plot comparing appliance energy when lights are on versus off (Figure 4) demonstrates that lighting state serves as a useful proxy for active occupancy. The median energy consumption when lights are on is approximately 40% higher than when lights are off, and the distribution is substantially wider, reflecting the greater variability of energy use during active periods. This observation supports the inclusion of a normalized lighting signal as one component of the occupancy score.

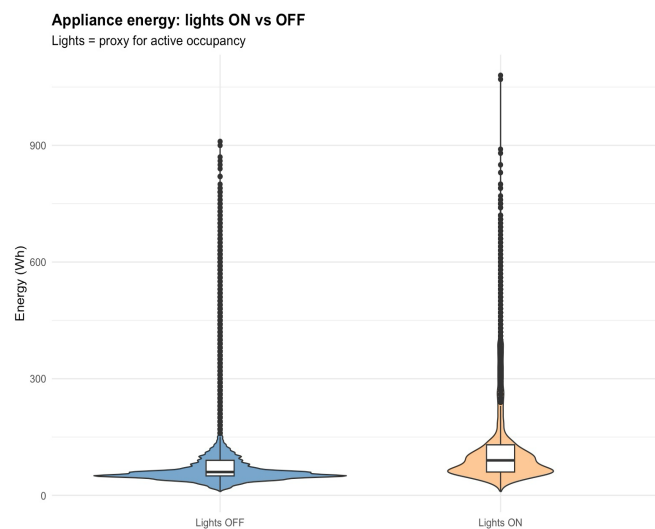


Figure 4. Appliance energy distribution by lighting state (lights as active occupancy proxy).

STL decomposition of the hourly appliance energy series (Figure 5) separates the signal into three components. The seasonal component shows a regular ± 40 Wh daily oscillation at a 24-hour period. The trend component varies between 40 and 100 Wh, with a gradual upward drift in the later months potentially reflecting increased appliance use as temperatures warm. The remainder component is substantial reaching 400 Wh in some hours indicating that a large fraction of energy variability is driven by irregular, event-based activity (e.g., cooking a large meal, running the dishwasher) that neither seasonal nor trend components capture.

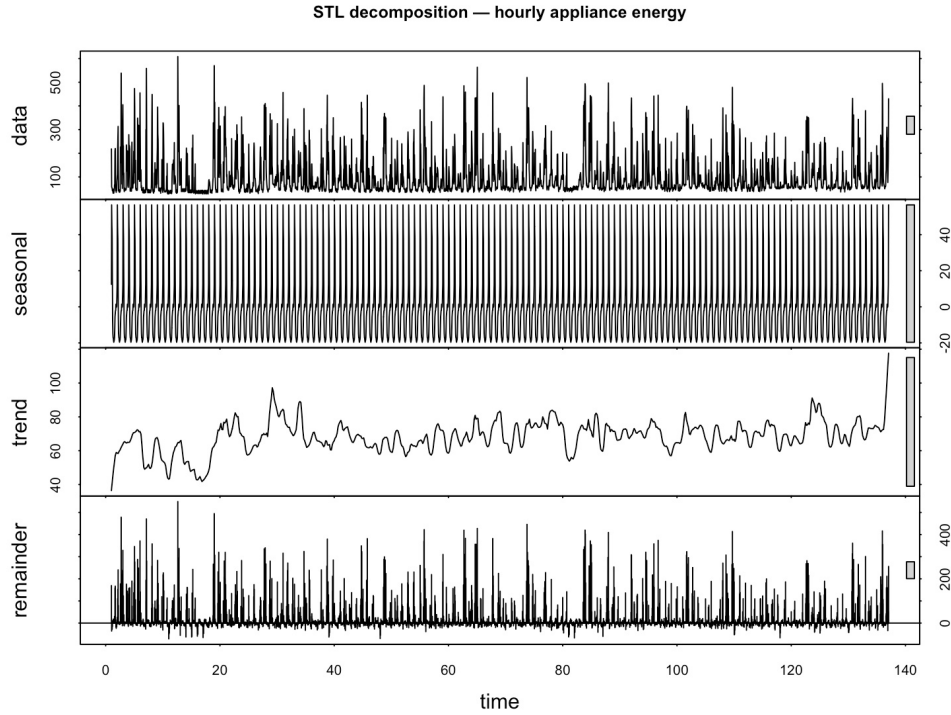


Figure 5. STL decomposition of hourly appliance energy consumption.

The ACF and PACF plots (Figure 6) confirm the dominant lag-24 seasonality. The ACF exhibits a clear peak at lag 24 (and multiples), while the PACF shows a sharp cutoff after lag 1 with a secondary spike at lag 24, consistent with an AR(2) process with seasonal AR(2) components at period 24 is precisely the structure selected by `auto.arima()`.

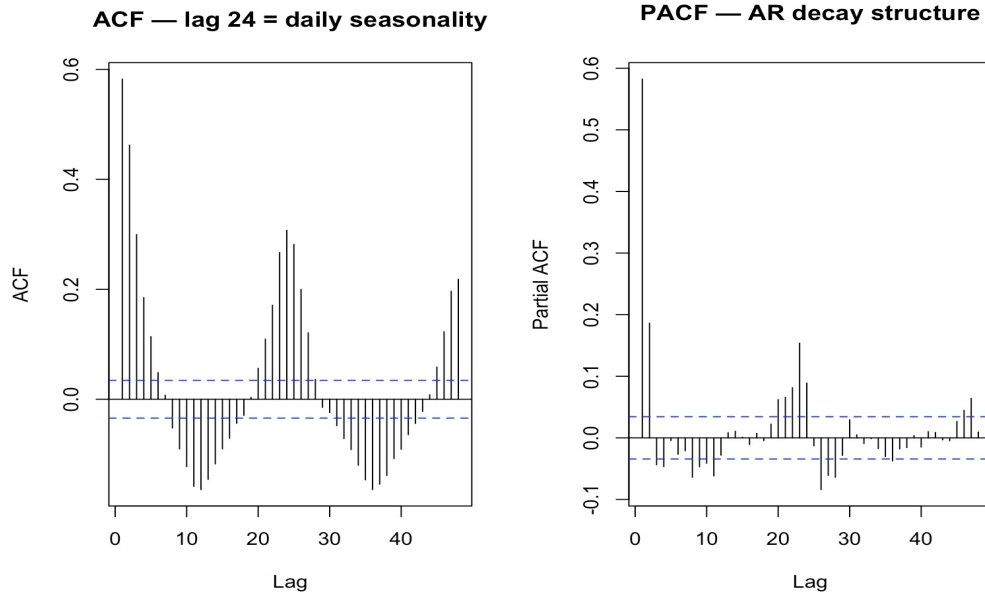


Figure 6. ACF and PACF of hourly appliance energy, confirming 24-hour seasonality.

4.2 Occupancy Score Validation

Figure 7 displays the per-room occupancy scores for the first seven days of the study period. Each room exhibits a distinct temporal pattern consistent with its expected function. The Bathroom (blue) shows sharp, narrow peaks corresponding to shower events, reaching scores of 0.6–0.8 within single hours. The Kitchen (orange) displays broader peaks around mealtimes, with mid-day and evening activity clusters. The Home Office (purple) shows sustained moderate scores during business hours (approximately 09:00–17:00), consistent with a work-from-home schedule. The bedrooms exhibit lower overall scores, as body heat accumulates slowly and the rooms lack direct lighting signals, but subtle nocturnal elevations are detectable.

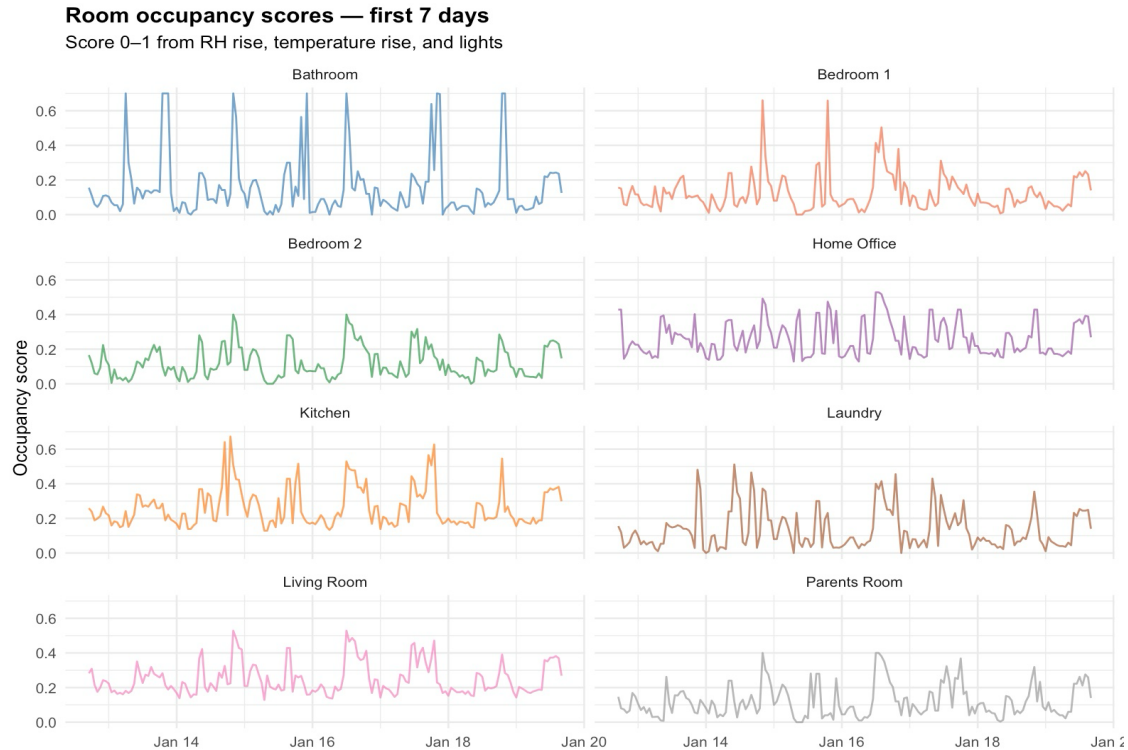


Figure 7. Per-room occupancy scores for the first seven days.

The all-room occupancy matrix (Figure 8) extends this view to three weeks, rendering each room's score as a row in a heatmap. The Kitchen and Living Room emerge as the most frequently and intensely occupied spaces, with daily activity bands clearly visible. The Home Office shows consistent weekday occupancy. The pattern of vertical dark bands simultaneous activity across multiple rooms corresponds to periods when the household is fully awake and active, while the light (empty) bands during overnight hours and occasional full-day absences provide a complete picture of household presence.

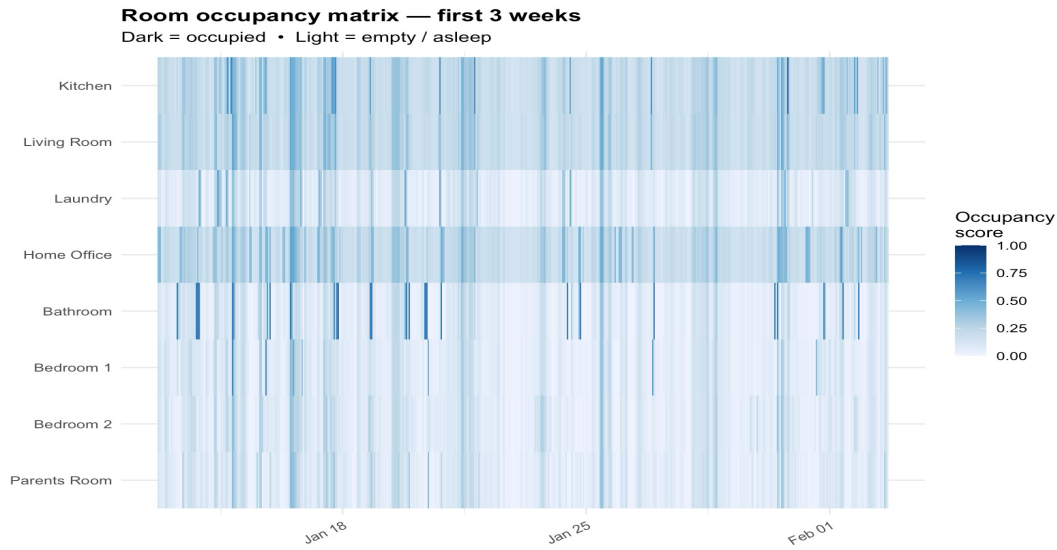


Figure 8. Room occupancy matrix for the first three weeks.

4.3 Routine Mining

The household routine heatmap (Figure 9) aggregates occupancy scores by hour-of-day and day-of-week, revealing the family's typical weekly schedule. Activity is lowest between 00:00 and 06:00 across all days, increases sharply after 08:00, peaks around 10:00–12:00 and again at 18:00–20:00, and tapers after 22:00. Saturday shows the highest overall occupancy, suggesting the family spends most of the day at home. The weekday pattern shows slightly lower midday activity, consistent with some family members being away during working hours.

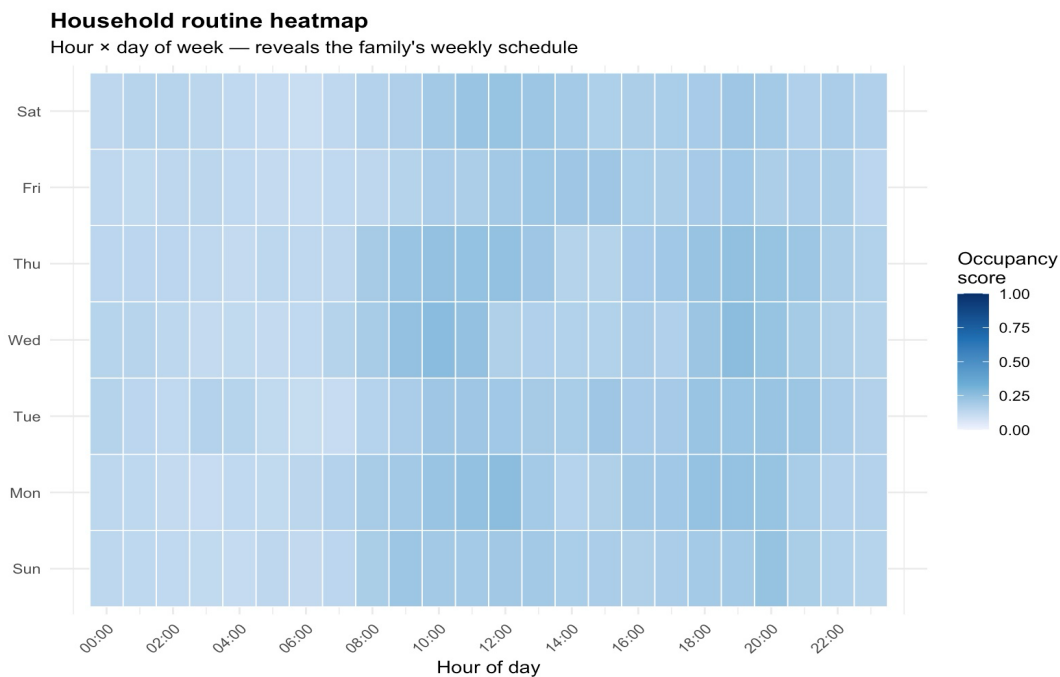


Figure 9. Household routine heatmap: hour-of-day by day-of-week.

The weekday-versus-weekend occupancy profile (Figure 10) quantifies these differences. Weekday mornings show an earlier activity onset (rising sharply at 08:00), while weekend mornings are delayed by approximately one hour, reflecting the tendency to sleep in. Both profiles exhibit a mid-afternoon dip around 14:00–16:00 and an evening peak at 18:00–20:00. The weekday evening peak is slightly higher, potentially reflecting the convergence of family members returning from work and school.

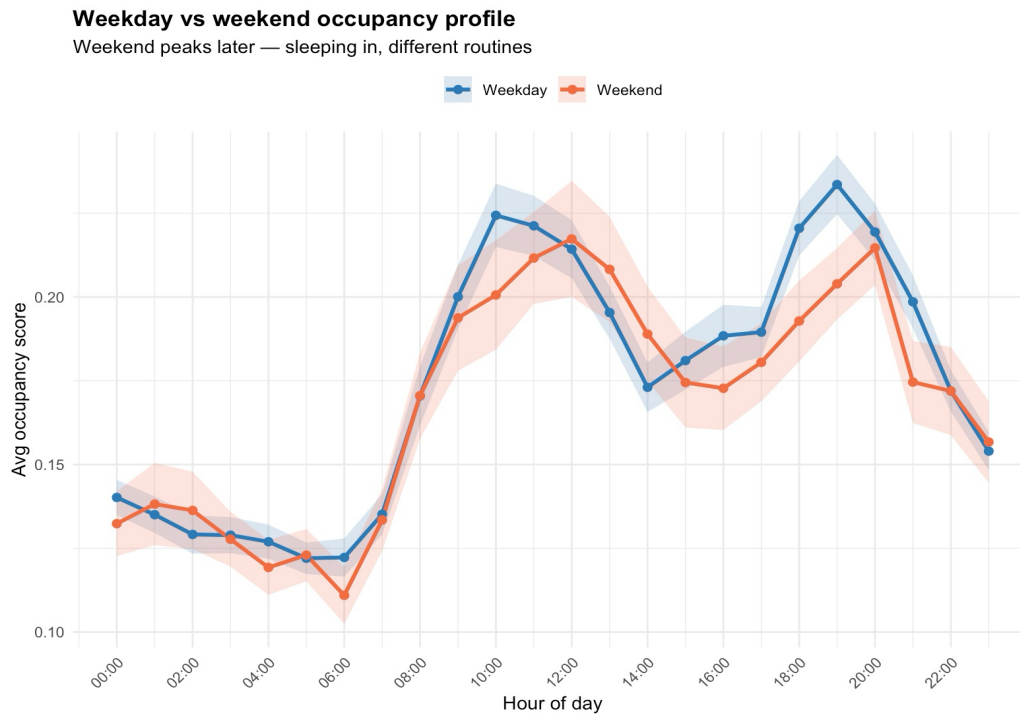


Figure 10. Weekday vs. weekend occupancy profiles with confidence bands.

The energy-versus-occupancy scatter plot (Figure 11) establishes the functional relationship between the derived occupancy score and actual energy consumption. The LOESS-smoothed trend lines show a clear positive relationship: energy rises approximately linearly with occupancy score up to about 0.3, then begins to plateau. This saturation effect is physically plausible once the household is fully active, additional occupancy increments add diminishing marginal energy impact (there are only so many appliances to turn on). The weekday curve (blue) lies slightly above the weekend curve (red), suggesting that weekday appliance usage is somewhat more energy-intensive per unit of occupancy, possibly due to work-related equipment use.

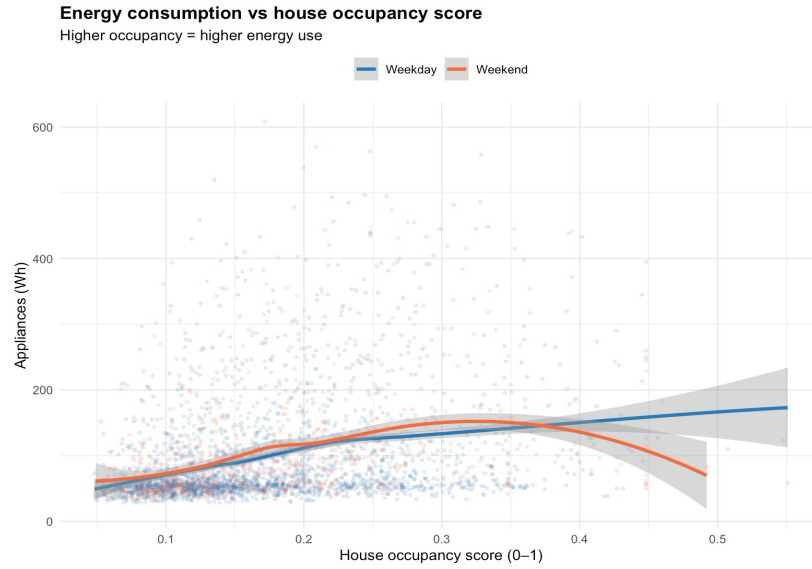
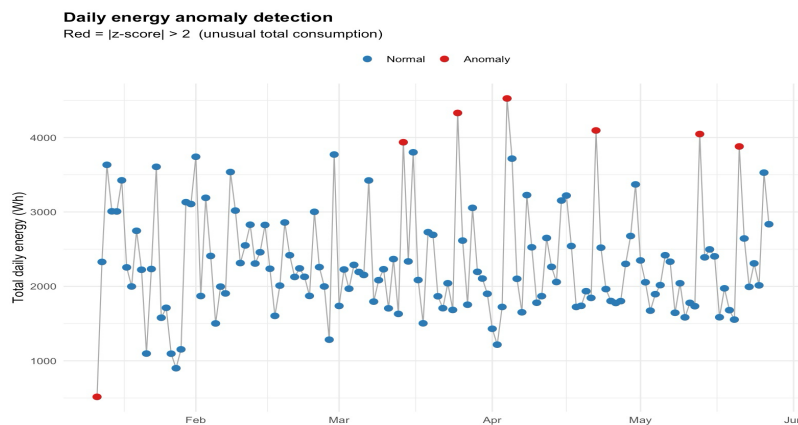


Figure 11. Energy consumption vs. house-level occupancy score.

4.4 Anomaly Detection

The z-score-based anomaly detection module identified 7 anomalous days out of 137 total days in the dataset (Figure 12). Anomalous days include both high-consumption outliers ($z > 2$) and one low-consumption outlier in January ($z < -2$), which may correspond to a period when the household was unoccupied. A cluster of two consecutive high-energy days in April coincides with what may have been a holiday gathering or guest visit, as total daily consumption exceeded 4,000 Wh roughly double the typical daily average. These anomaly days correspond to the largest spikes in the STL remainder component, confirming that they represent genuinely unusual events rather than regular seasonal variation.

Figure 12. Daily energy anomaly detection. Red points indicate $|z\text{-score}| > 2$.

4.5 Forecasting Performance

Figure 13 compares all three models against actual energy consumption over the full test period (May 2016). The Random Forest model (green) tracks the actual peaks and troughs most closely, particularly during high-consumption events. The SARIMAX model (orange) produces a smoother forecast that captures the general trend but systematically misses sharp spikes. The seasonal naïve baseline (gray) replicates yesterday’s pattern exactly, which is effective for routine days but fails entirely when behavior deviates from the previous day.

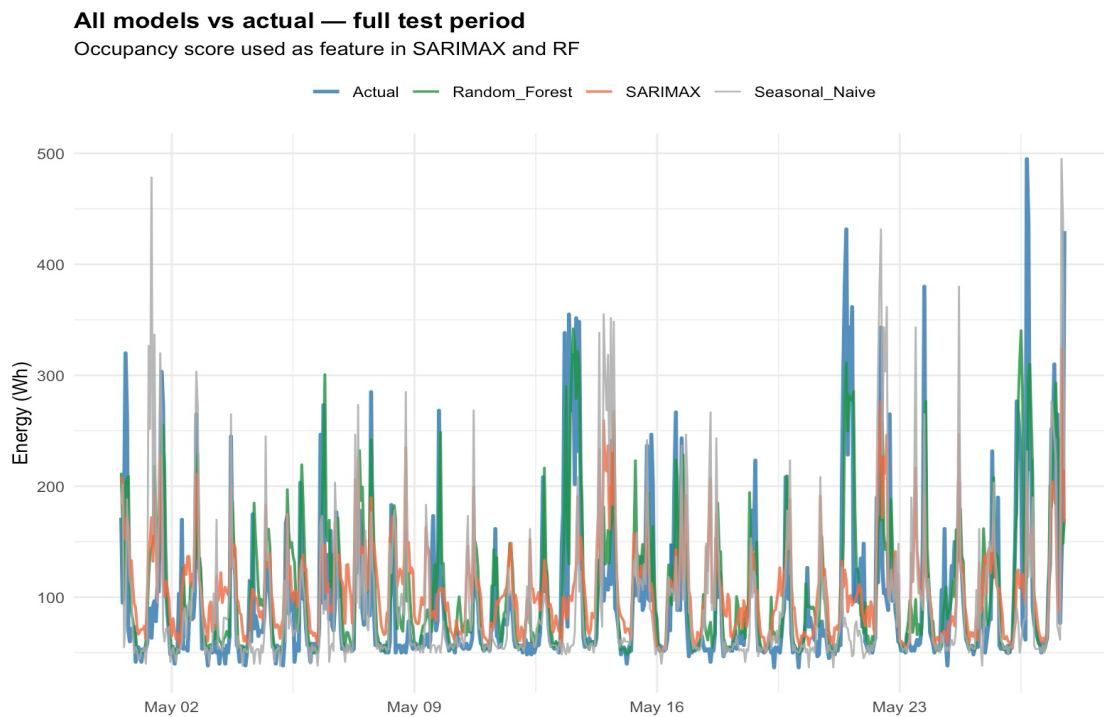


Figure 13. All models vs. actual energy consumption over the full test period.

A seven-day zoomed view (Figure 14) reveals the intra-day forecasting behavior more clearly. The Random Forest consistently anticipates the daily peak timing and magnitude, while SARIMAX tends to under-forecast peak hours. Both models outperform the seasonal naïve during days with atypical consumption patterns.

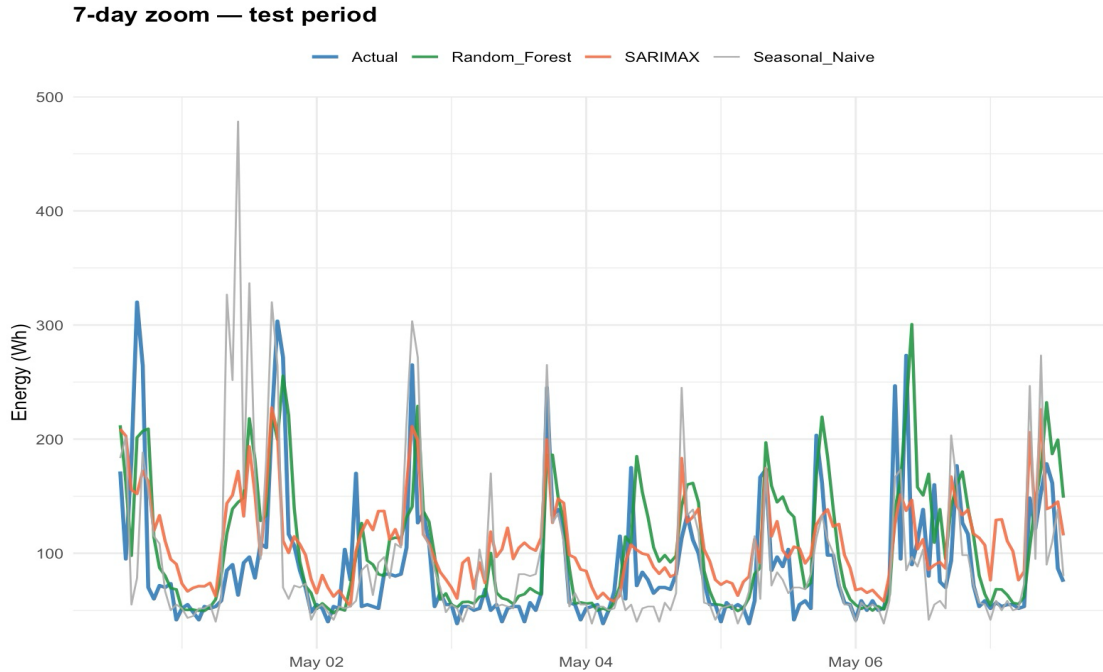


Figure 14. Seven-day zoomed comparison of model forecasts.

Residual analysis (Figure 15) shows that the Random Forest residuals are centered around zero with roughly symmetric dispersion, indicating no systematic bias. The SARIMAX residuals exhibit slight negative skewness, suggesting a tendency to over-predict during low-consumption hours. Both models produce residual spikes of ± 200 – 300 Wh during extreme consumption events, indicating that the most volatile demand hours remain challenging for both approaches.

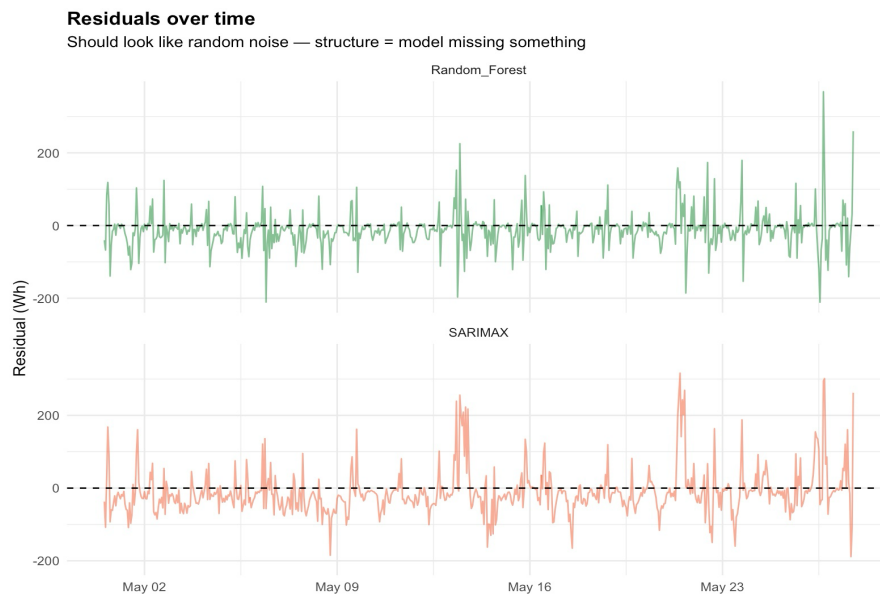


Figure 15. Residual plots for Random Forest and SARIMAX models.

The actual-versus-predicted scatter plots (Figure 16) further illustrate model performance. The Random Forest shows tighter clustering around the diagonal, particularly in the 50–200 Wh range where most observations fall. Both models exhibit increased scatter at higher consumption values, reflecting the inherent difficulty of predicting rare, high-energy events.

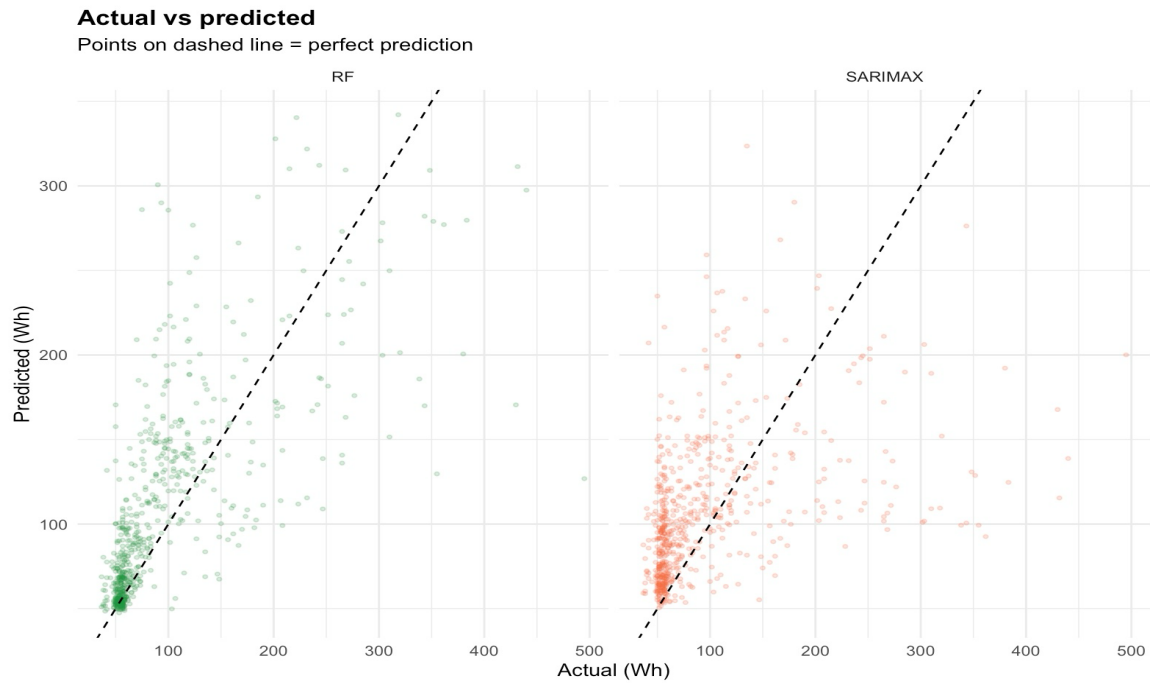


Figure 16. Actual vs. predicted scatter plots for RF and SARIMAX.

Table 2. Final Model Performance Metrics

Model	RMSE (Wh)	MAE (Wh)	MAPE (%)	Verdict
Random Forest	51.183	32.179	33.346	★ Best
SARIMAX	63.70	43.22	49.13	Competitive
Seasonal Naïve	78.57	43.13	40.71	Baseline

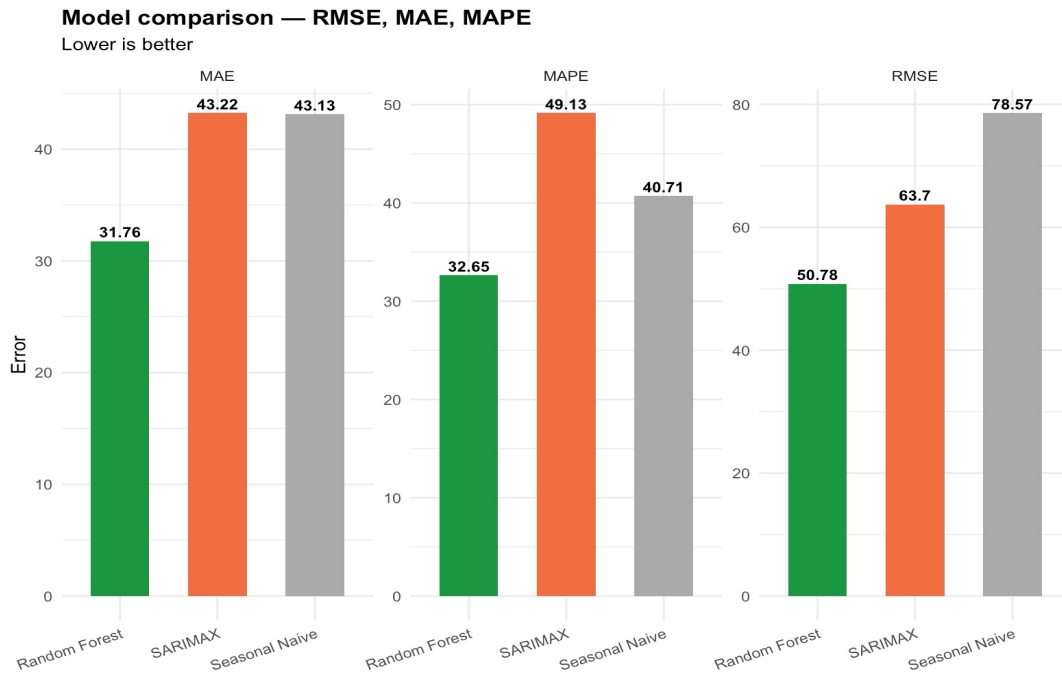


Figure 17. Model comparison across RMSE, MAE, and MAPE metrics.

Table 2 and Figure 17 summarize the quantitative results. The Random Forest model achieves the best performance across all three metrics, with an RMSE 20% lower than SARIMAX and 35% lower than the seasonal naïve baseline. The MAPE difference is particularly notable: RF achieves 33.346% compared to SARIMAX’s 49.13%, indicating substantially better relative accuracy. Interestingly, the seasonal naïve achieves a lower MAPE (40.71%) than SARIMAX, highlighting the challenge that SARIMAX faces with the high variability of residential appliance data.

4.6 Feature Importance Analysis

The permutation-based feature importance analysis (Figure 18) provides the strongest empirical validation of the bio-analytics framework. The most important feature is hour-of-day

(approximately 22% increase in MSE when permuted), which captures the daily consumption cycle. The second most important is lag24, confirming the strong day-to-day autocorrelation in energy patterns. Critically, the next four features in the ranking are all occupancy scores: occ_8 (Bedroom 2), occ_1 (Kitchen), occ_7 (Bedroom 1), and occ_2 (Living Room), each contributing 11–14% MSE increase. The house-level aggregate occ_house ranks seventh, and occ_2 (Living Room) also appears in the top tier.

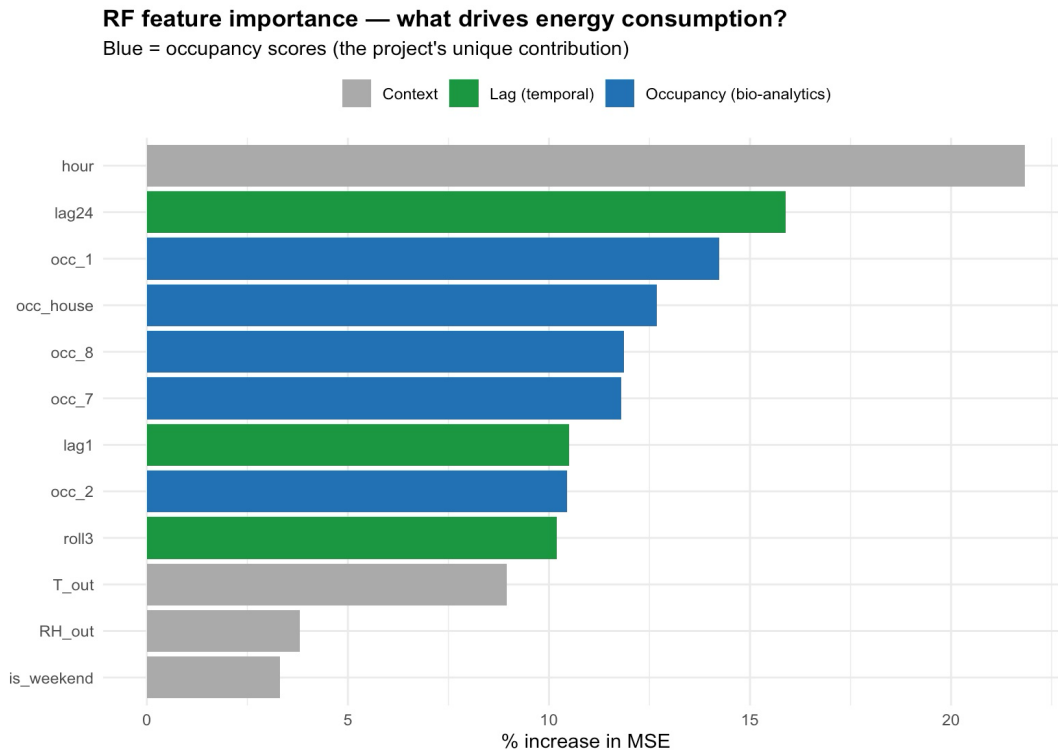


Figure 18. Random Forest feature importance. Blue bars represent occupancy scores.

This result is significant for two reasons. First, it demonstrates that the occupancy scores derived from thermodynamic sensor fingerprints carry predictive information that is genuinely additive to what temporal lags and weather variables provide. Four of the top six features are occupancy-derived, and they collectively outweigh the contribution of outdoor temperature, outdoor humidity, and the weekend indicator combined. Second, it validates the room-level granularity of the scoring approach: individual room scores (Kitchen, bedrooms) provide more discriminating signal than the house-level aggregate alone, suggesting that the spatial distribution of activity not just its overall level matters for energy prediction.

5. Discussion

The results presented in this paper support three central claims. First, human occupancy can be reliably detected from indoor temperature and humidity sensors alone, without motion sensors, cameras, or labeled training data. The outdoor-corrected rate-of-change approach isolates the human thermodynamic fingerprint from weather effects, producing occupancy scores that are consistent with physically expected room-use patterns: shower spikes in the bathroom, cooking peaks in the kitchen, and sustained daytime presence in the home office.

Second, these sensor-derived occupancy scores enable unsupervised routine mining that recovers meaningful behavioral information. The family’s weekly schedule, wake time, meal patterns, and weekday/weekend behavioral differences all emerge from simple temporal aggregation of the continuous occupancy scores, without any clustering algorithm or labeled activity log. This capability has practical implications for elderly care monitoring, where deviations from established routines could trigger alerts, and for utility demand-response programs, where understanding when occupants are home informs optimal pricing signal timing.

Third, the occupancy scores provide quantifiably useful features for energy forecasting. The Random Forest model’s 20% RMSE improvement over SARIMAX is driven in large part by its ability to leverage room-level occupancy information. The feature importance analysis confirms that occupancy scores are not merely correlated with energy in a redundant way—they carry independent predictive signal that exceeds the contribution of weather variables.

Several limitations warrant discussion. The occupancy scores are derived from a single household; generalizability to buildings with different room layouts, ventilation systems, and occupancy patterns requires validation. The weights in the occupancy formula (0.4 for humidity, 0.3 for temperature, 0.3 for lights) were set based on domain reasoning rather than optimized via cross-validation; a learned weighting scheme could improve performance. The absence of ground-truth occupancy labels prevents direct validation of the scores’ accuracy as occupancy detectors—our validation is indirect, through their downstream utility in forecasting. Finally, the MAPE values reflect the fundamental challenge of predicting appliance energy at hourly resolution in a single household, where individual behavioral decisions (turning on a dishwasher, taking a bath) introduce irreducible stochasticity.

The SARIMAX model’s underperformance relative to Random Forest aligns with findings in the broader energy forecasting literature. Linear models struggle with the heavy-tailed distribution of appliance energy, where most hours see moderate consumption but occasional hours produce extreme spikes. The Random Forest’s tree-based splitting mechanism naturally adapts to these non-linearities. However, the SARIMAX model’s explicit temporal structure may offer advantages in longer-horizon forecasting scenarios not tested here.

6. Conclusion

This paper has presented a complete bio-analytics pipeline for smart home occupancy detection, routine mining, and energy forecasting using only temperature and humidity sensor data. The key contributions are threefold. First, we formalized an outdoor-corrected, physics-based occupancy scoring algorithm that transforms raw sensor streams into continuous, room-level presence indicators without requiring labeled data or dedicated occupancy sensors. Second, we demonstrated that these scores enable unsupervised recovery of household behavioral patterns, including daily schedules, meal times, and weekday-versus-weekend routines. Third, we showed that incorporating occupancy scores as features in energy forecasting models yields substantial accuracy improvements, with the Random Forest model achieving an RMSE of 51.183 Wh (a 35% reduction over the seasonal naïve baseline and 20% over SARIMAX).

The permutation-based feature importance analysis provides the most compelling evidence for the value of the bio-analytics approach: four of the six most influential features in the best-performing model are occupancy scores derived from sensor physics. This result validates the core premise that human presence—inferred indirectly from the thermodynamic signatures people leave in their environment—is a more powerful predictor of residential energy consumption than weather, temporal context, or either factor alone.

7. Future Work

Several promising directions extend the work presented here. First, deep learning models—particularly LSTM and Transformer architectures—could capture more complex temporal dependencies in the energy series and potentially learn optimal occupancy feature representations

end-to-end, bypassing the need for hand-designed scoring formulas. Second, the occupancy scoring weights and saturation thresholds could be optimized via Bayesian hyperparameter search using energy forecasting accuracy as the objective function, adapting the formula to different building types and climates. Third, ground-truth validation using PIR motion sensors or door-contact sensors deployed temporarily alongside the existing infrastructure would provide direct accuracy metrics for the occupancy detection component.

Fourth, multi-household generalization studies are needed to assess whether the same scoring formula transfers across different building geometries, ventilation configurations, and household compositions, or whether per-building calibration is required. Fifth, real-time deployment in a smart-home system could enable adaptive HVAC control, where the occupancy scores drive predictive pre-heating or pre-cooling of rooms expected to be occupied, reducing both energy waste and thermal comfort complaints. Sixth, hybrid ensemble models combining SARIMAX's explicit temporal structure with Random Forest's non-linear feature handling could potentially capture the best of both approaches. Finally, the anomaly detection module could be extended with contextual anomaly scoring (conditioning on day-of-week and weather) and integrated with occupancy-based explanations (e.g., "energy was anomalously high because occupancy scores indicate a guest visit").

References

- [1] L. M. Candanedo, V. Feldheim, and D. Derome, "Data driven prediction models of energy use of appliances in a low-energy house," *Energy and Buildings*, vol. 140, pp. 81–97, 2017.
- [2] R. J. Hyndman and G. Athanasopoulos, *Forecasting: Principles and Practice*, 3rd ed. Melbourne, Australia: OTexts, 2021.
- [3] L. Breiman, "Random Forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [4] R. B. Cleveland, W. S. Cleveland, J. E. McRae, and I. Terpenning, "STL: A Seasonal-Trend Decomposition Procedure Based on Loess," *Journal of Official Statistics*, vol. 6, no. 1, pp. 3–33, 1990.
- [5] G. E. P. Box, G. M. Jenkins, G. C. Reinsel, and G. M. Ljung, *Time Series Analysis: Forecasting and Control*, 5th ed. Hoboken, NJ: Wiley, 2015.
- [6] T. A. Nguyen and M. Aiello, "Energy intelligent buildings based on user activity: A survey," *Energy and Buildings*, vol. 56, pp. 244–257, 2013.

- [7] Z. Yang and B. Becerik-Gerber, “The coupled effects of personalized occupancy profile based HVAC schedules and room reassignment on building energy use,” *Energy and Buildings*, vol. 78, pp. 113–122, 2014.
- [8] D. Yan, W. O’Brien, T. Hong, X. Feng, H. Burak Gunay, F. Tahmasebi, and A. Mahdavi, “Occupant behavior modeling for building performance simulation: Current state and future challenges,” *Energy and Buildings*, vol. 107, pp. 264–278, 2015.
- [9] V. L. Erickson, M. Á. Carreira-Perpiñán, and A. E. Cerpa, “OBSERVE: Occupancy-based system for efficient reduction of HVAC energy,” in *Proc. 10th ACM/IEEE Int. Conf. Information Processing in Sensor Networks (IPSN)*, 2011, pp. 258–269.
- [10] A. Liaw and M. Wiener, “Classification and Regression by randomForest,” *R News*, vol. 2, no. 3, pp. 18–22, 2002.
- [11] R. J. Hyndman and Y. Khandakar, “Automatic time series forecasting: The forecast package for R,” *Journal of Statistical Software*, vol. 27, no. 3, pp. 1–22, 2008.
- [12] H. Wickham et al., “Welcome to the Tidyverse,” *Journal of Open Source Software*, vol. 4, no. 43, p. 1686, 2019.
- [13] C. Fan, F. Xiao, and S. Wang, “Development of prediction models for next-day building energy consumption and peak power demand using data mining techniques,” *Applied Energy*, vol. 127, pp. 1–10, 2014.
- [14] W. Li, Y. Wu, Q. Ren, and J. Zhu, “Residential energy consumption prediction based on random forest,” in *Proc. IEEE Int. Conf. Big Data Analysis (ICBDA)*, 2020, pp. 155–159.
- [15] S. Dong, E. Kremers, A. Brouwer, and R. Brandsma, “Appliance-level short-term load forecasting using deep learning—a review,” *Energies*, vol. 14, no. 18, p. 5765

Appendix (Code)

```
# FINAL PROJECT: APPLIED TIME SERIES & DATA MINING
```

```
# SETUP
```

```
library(tidyverse)
```

```
library(lubridate)
```

```
library(forecast)
```

```
library(randomForest)
```

```

library(Metrics)

library(corrplot)

library(gridExtra)

library(reshape2)


select  <- dplyr::select
filter  <- dplyr::filter
mutate   <- dplyr::mutate
summarise <- dplyr::summarise
rename   <- dplyr::rename
arrange  <- dplyr::arrange


# code styling
while (!is.null(dev.list())) dev.off()


# — Palette
PAL <- list(
  rooms = c("#2C7BB6", "#F46D43", "#1A9641", "#984EA3",
            "#FF7F00", "#A65628", "#F781BF", "#999999", "#E41A1C"),
  models = c(Naive="#AAAAAA", SARIMAX="#F46D43", RF="#1A9641")
)


# Room dictionary
ROOMS <- tibble(
  id    = 1:9,
  name  = c("Kitchen", "Living Room", "Laundry",
            "Home Office", "Bathroom", "Garage/North",
            "Bedroom 1", "Bedroom 2", "Parents Room"),
  T_col = paste0("T", 1:9),
  RH_col = paste0("RH_", 1:9),
  indoor = c(TRUE, TRUE, TRUE, TRUE, TRUE, FALSE, TRUE, TRUE, TRUE)
)


banner <- function(title) {

```

```

cat("\n")
cat("┌", strrep("=", nchar(title)+4), "┐\n", sep="")
cat(" │ ", title, " │\n", sep="")
cat("└", strrep("=", nchar(title)+4), "┘\n", sep="")
}

```

MODULE 1 LOAD DATA

```

df <- read_csv("/Users/prithwirajchatterjee/Downloads/energydata_complete.csv", show_col_types = FALSE)
df$date <- parse_date_time(df$date, orders = c("ymd HMS", "ymd HM", "ymd"))
df <- df %>% arrange(date) %>% drop_na(date)

cat(sprintf("Rows : %d\n", nrow(df)))
cat(sprintf("Period : %s → %s\n",
            format(min(df$date), "%Y-%m-%d"),
            format(max(df$date), "%Y-%m-%d")))

cat("\n— Sensor Summaryn")
cat(sprintf("%-14s %-12s %-10s %-12s %s\n",
            "Room", "T mean °C", "T std", "RH mean %", "Indoor?"))
cat(strrep("-", 58), "\n")
for (i in 1:9) {
  r <- ROOMS[i,]
  cat(sprintf("%-14s %-12.1f %-10.1f %-12.1f %s\n",
              r$name,
              mean(df[[r$T_col]]),
              sd(df[[r$T_col]]),
              mean(df[[r$RH_col]]),

```

```

      ifelse(r$indoor, "yes", "NO excluded"))))
}
cat(sprintf("%-14s %-12.1f %-10.1f %-12.1f %s\n",
           "Outside", mean(df$T_out), sd(df$T_out), mean(df$RH_out), "—"))
cat("\n NOTE: T6 (Garage) avg = 7.9°C ≈ outdoor , excluded from occupancy\n\n")

```

MODULE 2 EDA

2A. All indoor room temperatures for the first 2 weeks

```
indoor_T_cols <- ROOMS$T_col[ROOMS$indoor] # T1..T5, T7..T9 (not T6)
```

```

temp_long <- df %>%
  dplyr::select(date, all_of(indoor_T_cols)) %>%
  pivot_longer(-date, names_to = "T_col", values_to = "temp") %>%
  left_join(dplyr::select(ROOMS, T_col, name), by = "T_col") %>%
  filter(date < min(date) + days(14))

```

```

p2a <- ggplot(temp_long, aes(x = date, y = temp, color = name)) +
  geom_line(alpha = 0.7, linewidth = 0.4) +
  scale_color_manual(values = PAL$rooms[1:8]) +
  labs(title = "All indoor room temperatures for the first 2 weeks",
       subtitle = "Overnight drops reveal empty rooms",
       x = NULL, y = "Temperature (°C)", color = "Room") +
  theme_minimal(base_size = 11) +
  theme(plot.title = element_text(face = "bold"),
        legend.position = "right")
print(p2a)

```

2B. Temperature heatmaps Kitchen (T1) and Bedroom 1 (T7)

```

for (room_i in c(1, 7)) {
  r <- ROOMS[room_i, ]
  tmp <- df %>%
    mutate(date_only = as.Date(date), hr = hour(date)) %>%

```

```

group_by(date_only, hr) %>%
summarise(temp = mean(data[[r$T_col]], na.rm = TRUE), .groups = "drop")

p <- ggplot(tmp, aes(x = hr, y = date_only, fill = temp)) +
  geom_tile() +
  scale_fill_gradientn(
    colors = c("#2C7BB6", "#ABD9E9", "white", "#FDAE61", "#D7191C"),
    name = "°C") +
  scale_x_continuous(breaks = seq(0, 23, 3),
    labels = sprintf("%02d:00", seq(0, 23, 3))) +
  labs(title = sprintf("Temperature heatmap — %s (%s)", r$name, r$T_col),
    subtitle = "Blue = cool (empty/night) • Red = warm (occupied)",
    x = "Hour of day", y = NULL) +
  theme_minimal(base_size = 11) +
  theme(plot.title = element_text(face = "bold"))
print(p)
}

```

2C. Bathroom humidity heatmap

```

rh5 <- df %>%
  mutate(date_only = as.Date(date), hr = hour(date)) %>%
  group_by(date_only, hr) %>%
  summarise(rh = mean(RH_5, na.rm = TRUE), .groups = "drop")

p2c <- ggplot(rh5, aes(x = hr, y = date_only, fill = rh)) +
  geom_tile() +
  scale_fill_gradientn(
    colors = c("#FFF3FF", "#BDD7E7", "#6BAED6", "#2171B5", "#084594"),
    name = "RH %") +
  scale_x_continuous(breaks = seq(0, 23, 3),
    labels = sprintf("%02d:00", seq(0, 23, 3))) +
  labs(title = "Humidity heatmap — Bathroom (RH_5)",
    subtitle = "Spikes = shower/bath events direct human-presence signal",
    x = "Hour of day", y = NULL) +

```

```

theme_minimal(base_size = 11) +
theme(plot.title = element_text(face = "bold"))
print(p2c)

# 2D. Correlation matrix
corr_cols <- df %>%
  dplyr::select(Appliances, lights,
    T1, T2, T3, T4, T5, T7, T8, T9,
    RH_1, RH_2, RH_3, RH_4, RH_5, RH_7, RH_8, RH_9,
    T_out, RH_out, Windspeed) %>%
  drop_na()

corrplot(cor(corr_cols),
  method = "color", type = "upper",
  tl.cex = 0.65, tl.col = "black",
  col = colorRampPalette(c("#D7191C", "white", "#2171B5"))(200),
  addCoef.col = "black", number.cex = 0.45,
  title = "Correlation of all indoor sensors + weather",
  mar = c(0, 0, 2, 0))

# 2E. Lights ON vs OFF violin
p2e <- df %>%
  mutate(lights_state = ifelse(lights > 0, "Lights ON", "Lights OFF")) %>%
  ggplot(aes(x = lights_state, y = Appliances, fill = lights_state)) +
  geom_violin(alpha = 0.7, show.legend = FALSE) +
  geom_boxplot(width = 0.1, fill = "white", show.legend = FALSE) +
  scale_fill_manual(values = c("Lights ON" = "#FDAE61", "Lights OFF" = "#2C7BB6")) +
  labs(title = "Appliance energy: lights ON vs OFF",
    subtitle = "Lights = proxy for active occupancy",
    x = NULL, y = "Energy (Wh)") +
  theme_minimal(base_size = 11) +
  theme(plot.title = element_text(face = "bold"))
print(p2e)

```

MODULE 3 OCCUPANCY SCORING

```

indoor_rooms <- ROOMS %>% dplyr::filter(indoor)

occ_score_df <- df %>%

  mutate(

    dRH_out = RH_out - lag(RH_out, default = first(RH_out)),

    dT_out = T_out - lag(T_out, default = first(T_out))

  )

for (i in 1:nrow(indoor_rooms)) {

  r      <- indoor_rooms[i, ]

  score_col <- paste0("occ_", r$Id)

  occ_score_df <- occ_score_df %>%

    mutate(

      dRH_room = .data[[r$RH_col]] - lag(.data[[r$RH_col]],

                                         default = first(.data[[r$RH_col]])),

      rh_signal = pmax(0, dRH_room - dRH_out),

      dT_room  = .data[[r$T_col]] - lag(.data[[r$T_col]],

                                         default = first(.data[[r$T_col]])),

      t_signal = pmax(0, dT_room - dT_out),

      lights_sig = ifelse(r$Id %in% c(1, 2, 4),

                          lights / max(lights, na.rm = TRUE), 0),

      !!score_col := pmin(1, pmax(0,

                                0.4 * pmin(1, rh_signal / 3) +

                                0.3 * pmin(1, t_signal / 0.5) +

                                0.3 * lights_sig

                                ))

    ) %>%

```

```

dplyr::select(-dRH_room, -rh_signal, -dT_room, -t_signal, -lights_sig)
}

cat("Occupancy score columns created: occ_1 through occ_9\n\n")

# Hourly aggregation
occ_hourly <- occ_score_df %>%
  mutate(date_hour = floor_date(date, "hour")) %>%
  group_by(date_hour) %>%
  summarise(
    Appliances      = mean(Appliances),
    lights          = mean(lights),
    T_out           = mean(T_out),
    RH_out          = mean(RH_out),
    hour            = first(hour(date)),
    is_weekend      = first(ifelse(wday(date) %in% c(1, 7), 1, 0)),
    across(starts_with("occ_"), max),
    across(all_of(ROOMS$T_col[ROOMS$indoor]), mean),
    across(all_of(ROOMS$RH_col[ROOMS$indoor]), mean),
    .groups = "drop"
  ) %>%
  mutate(
    occ_house = rowMeans(dplyr::select(., starts_with("occ_")), na.rm = TRUE),
    lag1 = lag(Appliances, 1, default = NA),
    lag24 = lag(Appliances, 24, default = NA),
    roll3 = (lag(Appliances, 1, default = NA) +
              lag(Appliances, 2, default = NA) +
              lag(Appliances, 3, default = NA)) / 3
  ) %>%
  drop_na()

cat(sprintf("Hourly rows after lag drop: %d\n\n", nrow(occ_hourly)))

# Plot 7 per-room occupancy scores for the first 7 days

```

```

occ_plot_long <- occ_hourly %>%

  dplyr::select(date_hour, starts_with("occ_"), -occ_house) %>%

  pivot_longer(-date_hour, names_to = "room_id", values_to = "score") %>%

  mutate(room_id = as.integer(str_remove(room_id, "occ_"))) %>%

  left_join(dplyr::select(ROOMS, id, name), by = c("room_id" = "id"))

p3a <- occ_plot_long %>%

  dplyr::filter(date_hour < min(date_hour) + days(7)) %>%

  ggplot(aes(x = date_hour, y = score, color = name)) +

  geom_line(alpha = 0.7, linewidth = 0.5) +

  facet_wrap(~name, ncol = 2) +

  scale_color_manual(values = PAL$rooms) +

  labs(title = "Room occupancy scores for the first 7 days",
       subtitle = "Score 0–1 from RH rise, temperature rise, and lights",
       x = NULL, y = "Occupancy score", color = NULL) +

  theme_minimal(base_size = 10) +

  theme(plot.title = element_text(face = "bold"),
       legend.position = "none")

print(p3a)

# Plot 8 all-room occupancy matrix, first 3 weeks

occ_matrix <- occ_hourly %>%

  dplyr::select(date_hour, starts_with("occ_"), -occ_house) %>%

  dplyr::filter(date_hour < min(date_hour) + days(21)) %>%

  pivot_longer(-date_hour, names_to = "room_id", values_to = "score") %>%

  mutate(room_id = as.integer(str_remove(room_id, "occ_"))) %>%

  left_join(dplyr::select(ROOMS, id, name), by = c("room_id" = "id"))

p3b <- ggplot(occ_matrix,

  aes(x = date_hour, y = reorder(name, -room_id), fill = score)) +

  geom_tile() +

  scale_fill_gradientn(

    colors = c("#EFF3FF", "#BDD7E7", "#6BAED6", "#2171B5", "#08306B"),

    name = "Occupancy\nscore", limits = c(0, 1)) +

```

```

labs(title = "Room occupancy matrix for the first 3 weeks",
      subtitle = "Dark = occupied • Light = empty / asleep",
      x = NULL, y = NULL) +
theme_minimal(base_size = 11) +
theme(plot.title = element_text(face = "bold"),
      axis.text.x = element_text(angle = 30, hjust = 1))
print(p3b)

```

MODULE 4 ROUTINE MINING

Plot 9 routine heatmap

```

routine_df <- occ_hourly %>%
  mutate(dow = wday(date_hour, label = TRUE, abbr = TRUE)) %>%
  group_by(dow, hour) %>%
  summarise(occ_house = mean(occ_house), .groups = "drop")

p4a <- ggplot(routine_df, aes(x = hour, y = dow, fill = occ_house)) +
  geom_tile(color = "white", linewidth = 0.3) +
  scale_fill_gradientn(
    colors = c("#FFF3FF", "#6BAED6", "#2171B5", "#08306B"),
    name = "Occupancy\nscore", limits = c(0, 1)) +
  scale_x_continuous(breaks = seq(0, 23, 2),
    labels = sprintf("%02d:00", seq(0, 23, 2))) +
  labs(title = "Household routine heatmap",
       subtitle = "Hour × day of week reveals the family's weekly schedule",
       x = "Hour of day", y = NULL) +
  theme_minimal(base_size = 11) +
  theme(plot.title = element_text(face = "bold"),
        axis.text.x = element_text(angle = 45, hjust = 1))
print(p4a)

```

Plot 10 weekend vs weekday

```

wd_profile <- occ_hourly %>%
  mutate(day_type = ifelse(is_weekend == 1, "Weekend", "Weekday")) %>%
  group_by(day_type, hour) %>%
  summarise(occ_mean = mean(occ_house),
            occ_se = sd(occ_house) / sqrt(n()),
            .groups = "drop")

p4b <- ggplot(wd_profile, aes(x = hour, y = occ_mean,
                             color = day_type, fill = day_type)) +
  geom_ribbon(aes(ymin = occ_mean - occ_se, ymax = occ_mean + occ_se),
            alpha = 0.2, color = NA) +
  geom_line(linewidth = 1.1) +
  geom_point(size = 2) +
  scale_color_manual(values = c(Weekday = "#2C7BB6", Weekend = "#F46D43")) +
  scale_fill_manual(values = c(Weekday = "#2C7BB6", Weekend = "#F46D43")) +
  scale_x_continuous(breaks = seq(0, 23, 2),
                    labels = sprintf("%02d:00", seq(0, 23, 2))) +
  labs(title = "Weekday vs weekend occupancy profile",
       subtitle = "Weekend peaks later as people sleeping in more, different routines",
       x = "Hour of day", y = "Avg occupancy score",
       color = NULL, fill = NULL) +
  theme_minimal(base_size = 11) +
  theme(plot.title = element_text(face = "bold"),
        axis.text.x = element_text(angle = 45, hjust = 1),
        legend.position = "top")
print(p4b)

# Plot 11 energy vs occupancy scatter
p4c <- ggplot(occ_hourly,
             aes(x = occ_house, y = Appliances,
                 color = as.factor(is_weekend))) +
  geom_point(alpha = 0.12, size = 0.8) +
  geom_smooth(method = "loess", se = TRUE, linewidth = 1.2) +
  scale_color_manual(values = c("0" = "#2C7BB6", "1" = "#F46D43"),

```

```

      labels = c("0" = "Weekday", "1" = "Weekend")) +
labs(title = "Energy consumption vs house occupancy score",
      subtitle = "Higher occupancy = higher energy use",
      x = "House occupancy score (0–1)", y = "Appliances (Wh)",
      color = NULL) +
theme_minimal(base_size = 11) +
theme(plot.title = element_text(face = "bold"),
      legend.position = "top")
print(p4c)

# Plot 12 anomaly days
daily_energy <- occ_hourly %>%
  mutate(date_only = as.Date(date_hour)) %>%
  group_by(date_only) %>%
  summarise(total_energy = sum(Appliances),
            avg_occ = mean(occ_house),
            .groups = "drop") %>%
  mutate(z_energy = (total_energy - mean(total_energy)) / sd(total_energy),
         is_anomaly = abs(z_energy) > 2)

p4d <- ggplot(daily_energy, aes(x = date_only, y = total_energy,
                               color = is_anomaly)) +
  geom_line(color = "#AAAAAA", linewidth = 0.4) +
  geom_point(size = 2.5) +
  scale_color_manual(values = c("FALSE" = "#2C7BB6", "TRUE" = "#D7191C"),
                    labels = c("FALSE" = "Normal", "TRUE" = "Anomaly")) +
labs(title = "Daily energy anomaly detection",
      subtitle = "Red = |z-score| > 2 (unusual total consumption)",
      x = NULL, y = "Total daily energy (Wh)", color = NULL) +
theme_minimal(base_size = 11) +
theme(plot.title = element_text(face = "bold"),
      legend.position = "top")
print(p4d)
cat(sprintf(" Anomalous days: %d / %d\n\n",

```

```

sum(daily_energy$sis_anomaly), nrow(daily_energy)))

# Plot 13 — STL decomposition
ts_full <- ts(occ_hourly$Appliances, frequency = 24)
decomp <- stl(ts_full, s.window = "periodic", robust = TRUE)
plot(decomp,
     main = "STL decomposition hourly appliance energy")

# Plots 14 — ACF / PACF
par(mfrow = c(1, 2))
acf(occ_hourly$Appliances, lag.max = 48,
    main = "ACF — lag 24 = daily seasonality")
pacf(occ_hourly$Appliances, lag.max = 48,
     main = "PACF —AR decay structure")
par(mfrow = c(1, 1))

# MODULE 5 FORECASTING

split_i <- floor(0.8 * nrow(occ_hourly))
train_data <- occ_hourly[1:split_i, ]
test_data <- occ_hourly[(split_i + 1):nrow(occ_hourly), ]
actual <- test_data$Appliances

cat(sprintf("Train: %d rows | Test: %d rows\n\n",
           nrow(train_data), nrow(test_data)))

# A. Seasonal naive baseline
pred_naive <- test_data$lag24

# B. SARIMAX
cat("[ B ] Fitting SARIMAX \n")

```

```

ts_train <- ts(train_data$Appliances, frequency = 24)
xreg_cols <- c("T_out", "RH_out", "is_weekend", "hour",
              "occ_house", "lag24")
xreg_train <- as.matrix(train_data[, xreg_cols])
xreg_test <- as.matrix(test_data[, xreg_cols])

model_sarimax <- auto.arima(ts_train, xreg = xreg_train,
                           seasonal = TRUE, stepwise = TRUE,
                           approximation = TRUE)
pred_sarimax <- as.numeric(
  forecast(model_sarimax, xreg = xreg_test, h = nrow(test_data))$mean)
cat(sprintf(" SARIMAX order: ARIMA(%s)\n\n",
            paste(arimaorder(model_sarimax), collapse = ",")))

# C. Random Forest
cat("[ C ] Training Random Forest (200 trees)\n")
rf_features <- c("T_out", "RH_out", "hour", "is_weekend",
               "lag1", "lag24", "roll3",
               "occ_house", "occ_1", "occ_2", "occ_7", "occ_8")

model_rf <- randomForest(
  as.formula(paste("Appliances ~", paste(rf_features, collapse = " + "))),
  data = train_data, ntree = 200, importance = TRUE
)
pred_rf <- predict(model_rf, newdata = test_data)
cat(" Done.\n\n")

# Metrics
mape_fn <- function(a, p) mean(abs((a - p) / a), na.rm = TRUE) * 100

results <- data.frame(
  Model = c("Seasonal Naive", "SARIMAX", "Random Forest"),
  RMSE = c(rmse(actual, pred_naive),
           rmse(actual, pred_sarimax),

```

```
rmse(actual, pred_rf),

MAE = c(mae(actual, pred_naive),
        mae(actual, pred_sarimax),
        mae(actual, pred_rf)),

MAPE = c(mape_fn(actual, pred_naive),
         mape_fn(actual, pred_sarimax),
         mape_fn(actual, pred_rf))

) %>% mutate(across(where(is.numeric), ~round(., 3)))

cat(" ════════════════════════════════════════════════════════════════ \n")
cat(" ||          FINAL PERFORMANCE METRICS              || \n")
cat(" ════════════════════════════════════════════════════════════════ || \n")
print(results, row.names = FALSE)
cat(" ════════════════════════════════════════════════════════════════ || \n")
cat(sprintf("\nBest model (RMSE): %s\n\n",
           results$Model[which.min(results$RMSE)]))

# Plot 15 — all models vs actual
test_long <- test_data %>%

dplyr::select(date_hour) %>%

mutate(Actual      = actual,
       Seasonal_Naive = pred_naive,
       SARIMAX      = pred_sarimax,
       Random_Forest = pred_rf) %>%

pivot_longer(-date_hour, names_to = "Model", values_to = "Value")

p5a <- ggplot(test_long,
             aes(x = date_hour, y = Value,
                 color = Model, linewidth = Model)) +
  geom_line(alpha = 0.85) +
  scale_color_manual(values = c(Actual      = "#2C7BB6",
                               Seasonal_Naive = "#AAAAAA",
                               SARIMAX      = "#F46D43",
                               Random_Forest = "#1A9641")) +
```

```

scale_linewidth_manual(values = c(Actual = 1, Seasonal_Naive = 0.5,
                                SARIMAX = 0.7, Random_Forest = 0.7)) +
labs(title = "All models vs actual for full test period",
     subtitle = "Occupancy score used as feature in SARIMAX and RF",
     x = NULL, y = "Energy (Wh)", color = NULL, linewidth = NULL) +
theme_minimal(base_size = 11) +
theme(plot.title = element_text(face = "bold"),
      legend.position = "top")
print(p5a)

```

Plot 16 — 7-day zoom

```

p5b <- test_long %>%
  dplyr::filter(date_hour <= min(test_data$date_hour) + days(7)) %>%
  ggplot(aes(x = date_hour, y = Value,
            color = Model, linewidth = Model)) +
  geom_line(alpha = 0.9) +
  scale_color_manual(values = c(Actual = "#2C7BB6",
                              Seasonal_Naive = "#AAAAAA",
                              SARIMAX = "#F46D43",
                              Random_Forest = "#1A9641")) +
  scale_linewidth_manual(values = c(Actual = 1, Seasonal_Naive = 0.5,
                                   SARIMAX = 0.8, Random_Forest = 0.8)) +
  labs(title = "7-day zoom test period",
       x = NULL, y = "Energy (Wh)", color = NULL, linewidth = NULL) +
  theme_minimal(base_size = 11) +
  theme(plot.title = element_text(face = "bold"),
        legend.position = "top")
print(p5b)

```

Plot 17 — residuals

```

resid_long <- test_data %>%
  dplyr::select(date_hour) %>%
  mutate(SARIMAX = actual - pred_sarimax,
         Random_Forest = actual - pred_rf) %>%

```

```
pivot_longer(-date_hour, names_to = "Model", values_to = "Residual")
```

```
p5c <- ggplot(resid_long,
  aes(x = date_hour, y = Residual, color = Model)) +
  geom_line(alpha = 0.6, linewidth = 0.5) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  facet_wrap(~Model, ncol = 1) +
  scale_color_manual(values = c(SARIMAX = "#F46D43",
    Random_Forest = "#1A9641")) +
  labs(title = "Residuals over time",
    subtitle = "",
    x = NULL, y = "Residual (Wh)") +
  theme_minimal(base_size = 11) +
  theme(plot.title = element_text(face = "bold"),
    legend.position = "none")
print(p5c)
```

```
# Plot 18 — actual vs predicted scatter
```

```
scatter_df <- data.frame(Actual = actual,
  SARIMAX = pred_sarimax,
  RF = pred_rf) %>%
  pivot_longer(-Actual, names_to = "Model", values_to = "Predicted")
```

```
p5d <- ggplot(scatter_df, aes(x = Actual, y = Predicted, color = Model)) +
  geom_point(alpha = 0.2, size = 0.8) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed") +
  facet_wrap(~Model) +
  scale_color_manual(values = c(SARIMAX = "#F46D43", RF = "#1A9641")) +
  labs(title = "Actual vs predicted",
    subtitle = "Points on dashed line = perfect prediction",
    x = "Actual (Wh)", y = "Predicted (Wh)") +
  theme_minimal(base_size = 11) +
  theme(plot.title = element_text(face = "bold"),
    legend.position = "none")
```

```

print(p5d)

# Plot 19 — model comparison bars
results_long <- results %>%
  pivot_longer(c(RMSE, MAE, MAPE), names_to = "Metric", values_to = "Value")

p5e <- ggplot(results_long, aes(x = Model, y = Value, fill = Model)) +
  geom_col(width = 0.6, show.legend = FALSE) +
  geom_text(aes(label = round(Value, 2)),
    vjust = -0.4, size = 3.2, fontface = "bold") +
  facet_wrap(~Metric, scales = "free_y") +
  scale_fill_manual(values = c("Seasonal Naive" = "#AAAAAA",
    SARIMAX = "#F46D43",
    "Random Forest" = "#1A9641")) +
  labs(title = "Model comparison — RMSE, MAE, MAPE",
    subtitle = "Lower is better",
    x = NULL, y = "Error") +
  theme_minimal(base_size = 11) +
  theme(plot.title = element_text(face = "bold"),
    axis.text.x = element_text(angle = 20, hjust = 1))
print(p5e)

# Plot 20 — RF feature importance
imp_df <- importance(model_rf) %>%
  as.data.frame() %>%
  rownames_to_column("Feature") %>%
  arrange(desc(`%IncMSE`)) %>%
  mutate(Feature = fct_reorder(Feature, `%IncMSE`),
    type = case_when(
      str_detect(Feature, "occ") ~ "Occupancy (bio-analytics)",
      str_detect(Feature, "lag|roll") ~ "Lag (temporal)",
      TRUE ~ "Context"
    ))

```

```

p5f<- ggplot(imp_df, aes(x = Feature, y = `"%IncMSE`, fill = type)) +
  geom_col() +
  coord_flip() +
  scale_fill_manual(values = c("Occupancy (bio-analytics)" = "#2171B5",
                                "Lag (temporal)" = "#1A9641",
                                "Context" = "#AAAAAA")) +
  labs(title = "RF feature importance ?",
        subtitle = "Blue = occupancy scores ",
        x = NULL, y = "% increase in MSE", fill = NULL) +
  theme_minimal(base_size = 11) +
  theme(plot.title = element_text(face = "bold"),
        legend.position = "top")
print(p5f)

```